

SCQ (Single Correct Type) :

1. We represent various reagents pictorially such that,



⇒ 0.1 M strong acid;



⇒ 0.1 M strong base



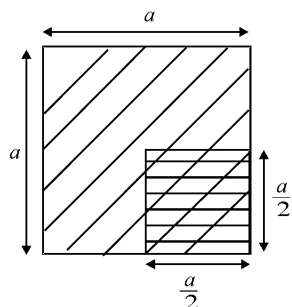
⇒ 0.1 M weak acid ($K_a = 10^{-5}$);



⇒ 0.1 M weak base ($K_b = 10^{-5}$)

The relative area of the squares represents the relative volume of the reagents used and the overlapping region represents the relative amount of mixing of the reagents. Consider the temperature to be 298 K.

(use $\log 5 = 0.7$, $\log 2 = 0.3$, $\log 3 = 0.5$, $d_{H_2} = 1 \text{ g ml}^{-1}$) For a solution represented by



Which is incorrect ?

- (A) $\text{pH} > 1$ (B) α of weak acid = 5×10^{-4}
(C) α of $\text{H}_2\text{O} = 9 \times 10^{-15}$ (D) $\text{pH} = 1.5$
2. For a homogeneous gaseous reaction $2\text{A}_{(g)} + 3\text{B}_{(g)} \rightleftharpoons 4\text{C}_{(g)} + \text{D}_{(g)}$ $K_{eq} = \frac{8}{(1.5)^3}$. If in a 2 litre rigid

container starting with 4 moles of A and 6 moles of B equilibrium was established then the option which is incorrect.

- (A) Concentration of B at equilibrium is 1.5 M (B) Concentration of D at equilibrium is 1 M
(C) Concentration of A at equilibrium is 1 M (D) Concentration of C at equilibrium is 2 M

MCQ (Multiple Correct Type) :

3. An ideal gas with adiabatic component undergoes a process in which internal energy depends upon volume as $U = aV^\alpha$ then select correct statements:

(A) Molar heat capacity of process is $\left(\frac{R}{\alpha} - \frac{R}{1-\gamma} \right)$

(B) Heat charge in process is given by $\Delta U \left[1 + \frac{(1-\gamma)}{\alpha} \right]$

- (C) Change in internal energy is $\frac{(1-\gamma)}{\alpha} \Delta T$
- (D) Equation of process is $PV^x = \text{constant}$, where $x > 0$ if $\alpha < 1$.
4. $\text{Ag(s)} | \text{AgBr(s)} | \text{Br}^- (0.01\text{M}) || \text{Cl}^- (0.2\text{M}) | \text{AgCl(s)} | \text{Ag(s)}$
 If K_{sp} of AgBr & AgCl at 25°C are 10^{-12} & 10^{-10} respectively find which is/are true for above cell at 25°C ($\log 2 = 0.3$).
- (A) As $E_{\text{Ag}^+/\text{Ag}}^0$ is not given so E_{Cell} can not be determined.
- (B) E_{Cell} is 0.0413 V (approx)
- (C) $\Delta G < 0$
- (D) $E^0 = 0$
5. Which of the following cannot be converted into anhydrous chloride only by heating?
- (A) $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ (B) $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ (C) $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ (D) $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$
6. The correct option(s) regarding the complex $[\text{Co(en)}(\text{NH}_3)_3(\text{H}_2\text{O})]^{3+}$ is/are :
- (A) It is diamagnetic
- (B) It has two geometrical isomers
- (C) It will have three geometrical isomers if bidentate 'en' replaced by two Cl^- ligands
- (D) It absorbs light at longer wavelength as compound to $[\text{Co(en)}(\text{NH}_3)_4]^{3+}$
7. Select correct statement(s).
- (A) Addition of high proportion of manganese makes steel useful in making rails of rail roads
- (B) Any Fe_2O_3 which escapes reduction in the "zone of reduction" is reduced in "zone of heat absorption" by carbon in the extractive metallurgy of iron.
- (C) Below reaction is an example of hydrometallurgy
 $\text{CuFeS}_2 + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{FeSO}_4 + 2\text{H}_2\text{S}$
- (D) All 3 minerals are named correctly
 (i) Pyrargyrite : Ag_3SbS_3 (ii) Cinnabar : HgS (iii) Felspar : $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$
8. $\text{FeSO}_4(\text{s}) \xrightarrow{\Delta} \text{Solid X} + \text{gas A} + \text{gas B}$
 $\text{X} + \text{KOH} + \text{Cl}_2 \rightarrow \text{Z} + \text{KC}_1 + \text{H}_2\text{O}$
 $\text{Z} + \text{H}_2\text{O}(\ell) \xrightarrow{\text{decomposes}} \text{Solid X} + \text{O}_2(\text{g}) + \text{Base}$
 $\text{gas A} + \text{K}_2\text{Cr}_2\text{O}_7 \longrightarrow \text{green solution}$
 Choose the correct statements
- (A) The trimeric form of gas B has no S-S bond (B) Z has Fe in +6 oxidation state
- (C) Gas A gives white turbidity with H_2S gas (D) None of these

9. In context to metallic crystal (metal atom A originally), which of the following is incorrect statement(s)?
(CN means coordination number, PF = packing fraction, a = edge length of unit cell)
(A) Number of nearest neighbour of A will be 12 if 'A' are in FCC unit cell
(B) If 'A' crystalizes in ideal BCC as well as FCC (at two diff Temperatures) then $(a_{\text{BCC}} / a_{\text{FCC}}) = (PF_{\text{BCC}}/PF_{\text{FCC}})$
(C) Total number of TV and OV are 8 and 4 respectively if A' are in FCC unit cell.
(D) 'A' atoms are in FCC, if all face centre atoms are replaced with 20% bigger metal atoms 'B', then A-A atoms remain in contact with each-other.
10. Yellow ppt. (W) $\xleftarrow[\text{H}^+]{\text{K}_2\text{CrO}_4}$ X $\xrightarrow{\text{Dil.HCl}}$ Y Yellow ppt. + Z [Pungent smelling gas]
X $\rightarrow$ Gives green colour with flame test.
(A) 'X' can be BaS_2O_3 with 'W' as BaCrO_4 (B) 'X' can be BaS_2O_3 with 'Y' as BaSO_4
(C) 'X' can be BaS_2O_3 with 'Z' as SO_2 (D) 'X' can be BaS_2O_3 with 'Y' as Sulphur
11. Choose the correct statement/s:
(A) BeCO_3 is kept in the atmosphere of CO_2 because it is less stable thermally
(B) Be dissolve in alkali forming $[\text{Be}(\text{OH}_4)]^{2-}$
(C) Hydrolysis of BF_3 is partial in nature.
(D) Mixture of CaC_2 and Ca_3P_2 is used as Holme's signals.

(Numerical Value type) :

12. A man weighing 80 kg was injected with 5 ml of water containing tritium giving 1.08×10^{11} dpm. When the sample is distributed uniformly within the body water of patient, a 1.0 ml sample of water showed an activity of 3.0×10^4 dps. Estimate the weight percentage of water in the human body. Neglect any change in volume of water in the human body during the observation period and take the density of water as 1000 kg/m^3 . (Half life of ${}^3\text{H}$ is 12.3 years).
[If answer is 82.45 % then fill OMR as 8245]

13. In Raymonds factory, H_2O_2 is used for bleaching clothes. For avoiding errors, bottles are prepared according to the type of cloth. Each bottle has volume 500 ml and only one bottle is used for 1000 gm cloth. Information about bottles is given in the following chart:

Type of cloth	Re quired mass of O_2 per 100 gm of cloth	Name of Bottle to bleach 1 kg cloth
Cotton	3.2gm	P
Polyster	4.8gm	Q
Silk	5.6 gm	R

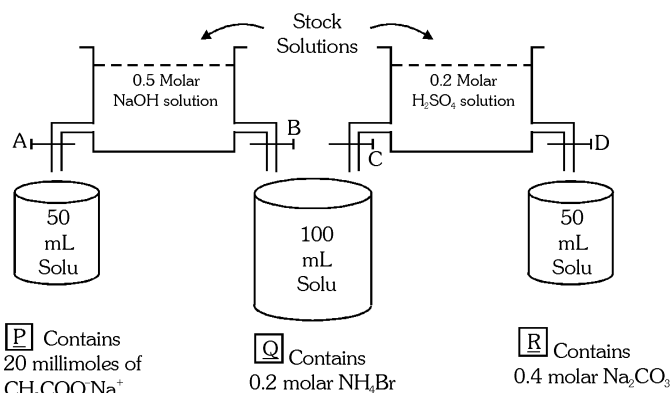
The molality of H_2O_2 solution in "Q" type Bottle is, if the density of solution is 2 gm/ml.

14. Number of following which are non planar species and having similar all bond lengths = X.

C1F_3 , NCl_3 , CH_3^- , ClO_2^- , BeCl_3^- , SO_3^{2-} , XeO_2F_2 , B_2H_6 , $[\text{PCl}_4]^+$, $\text{B}(\text{OH})_3$

Specify : $X \times 30$.

15.



If valves are opened sequentially (A to D) & each valve lets out 20 mL of stock solution, then how many statement/s is/are correct ? (Temp = 25°C)

- After opening (A), buffer is created in beaker "P".
 - After opening (B), buffer is created beaker "Q".
 - After opening (C), contents in beaker Q can still act as buffer even though solutions pH decreases.
 - After opening (D), solution of $\text{pH} < 7$ is created in beaker "R".
16. A sample of sea water was found to contain 9.5%w/w of MgCl_2 and 5.85% w/w of NaCl . Assuming complete ionisation of salts, what will be the boiling point of sample. (K_b for water = 0.52, at. wt. of Mg = 24, at. wt. of Na = 23, at. wt. of Cl = 35.5) :-
17. A white powder turns yellow (B) on heating and evolves a gas which turns lime water milky, as well as water vapour. The yellow residue turns white on cooling, but will turns yellow again when heated. B reacts with dilute sulphuric acid to give a colourless solution (C). If dilute NaOH is added to C, a white precipitate (D) is formed initially, which dissolves on adding excess base. With dilute ammonia solution, C gives a white precipitate which dissolves in excess ammonia, giving a clear solution (E). Identify A to E.
18. Complete the following reactions:
- $\text{Cl}_2\text{O} + \text{NaOH} \longrightarrow$
 - $\text{HClO}_4 + \text{P}_4\text{O}_{10} \longrightarrow$
 - $\text{ClO}_2 + \text{NaOH} \longrightarrow$
 - $\text{KBrO}_3 + \text{F}_2 + \text{KOH} \longrightarrow$
 - $\text{Cl}_2\text{O}_6 + \text{NaOH} \longrightarrow$
 - $\text{Cl}_2\text{O}_6 + \text{HF} \longrightarrow$
19. Complete the following chemical reactions:
- $\text{F}_2 + \text{H}_2\text{O} \longrightarrow$
 - $\text{Cl}_2 + \text{CO} \longrightarrow$
 - $\text{Cl}_2 + \text{NH}_3$ (excess) $\longrightarrow$
 - $\text{Cl}_2 + \text{NaOH} \longrightarrow$
 - $\text{SiO}_2 + \text{HF} \longrightarrow$

20. A 0.016M of an acid solution in benzene is dropped on a water surface, the benzene evaporates and the acid forms a monomolecular film of solid type. What volume of the above solution would be required to cover a 500 cm² surface area of water with monomolecular layer of acid ? Area covered by single acid molecule is 0.2 nm² .
21. The wave function for electron in ground state of hydrogen atom is $\psi_{1s} = \left(\pi a_0^3\right)^{-\frac{1}{2}} e^{-\frac{r}{a_0}}$, where “ a_0 ” is radius of Bohr’s orbit. Calculate the probability of finding the electron somewhere between 0 and $2a_0$.
22. To a 10mL 1.0 M aqueous solution of Br₂, excess of NaOH is added so that all Br₂ disproportionated to Br⁻ and BrO₃⁻ .The resulting solution is freed from bromide ion by extraction and excess of OH⁻ neutralized by acidifying the solution. The resulting solution is just sufficient to react with 1.5 g of an impure CaC₂O₄ sample. Calculate percentage purity of oxalate sample.

Answer key

1. (D) 2. (B) 3. (AD) 4. (BC) 5. (ABCD)
6. (ABCD) 7. (ABC) 8. (ABC) 9. (BD) 10. (ACD)
11. (ABD) 12. (7500) 13. (3.34) 14. (120) 15. (2)
16. (103.07) 17. (4)
18. A : $\text{ZnCO}_3 \cdot \text{Zn(OH)}_2$, B : ZnO , C : ZnSO_4 , D : Zn(OH)_2 , E : $[\text{Zn(NH}_3)_4]^{2+}$
19.
20.
21. $V = 25.94 \times 10^{-3} \text{ mL}$ 22. 18.48 min 23. 0.7624 24. 25. 85.33